

Notes: Drucker & Puri (RFS, 2009)

On Loan Sales, Loan Contracting, and Lending Relationships

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1 Big picture

- **Question.** What makes loans saleable in the secondary loan market, and what are the borrower-level consequences of loan sales for credit availability and lending relationships?
- **Core setting.** The paper studies corporate loans from 1999 to 2004, combining secondary-market loan quote data with DealScan loan contracts, borrower characteristics, credit ratings, and future lending relationships.
- **Main friction.** Loan sales create information and agency problems. Loan buyers may know less than originators, may monitor less effectively, and may worry that sellers are offloading bad loans.
- **Contracting mechanism.** The paper studies whether loan contracts use covenants to mitigate these problems. Covenants give buyers intervention rights and constrain borrowers, making traded loans easier to monitor.
- **Main contract result.** Sold loans have more covenants and tighter net-worth covenants. The effects are strongest when information and agency problems are more severe.
- **Borrower benefit.** Borrowers accept more restrictive contracts because loan sales appear to increase access to private debt capital. Sold loans are larger, and borrowers' debt rises around the year of the loan sale.
- **Relationship result.** Contrary to the concern that loan sales destroy relationship lending, borrowers whose loans are sold are more likely to borrow again from the same lead lender, especially among riskier borrowers.
- **Economic takeaway.** The secondary loan market does not simply replace relationship banking with transactional lending. Instead, restrictive contracts and risk sharing allow banks to originate risky, large loans while maintaining durable lending relationships.

2 Incremental contribution of the paper

- **Loan sales and contract design.** Earlier work studies why banks sell loans or whether sold-loan announcements matter. This paper adds contract-level evidence showing that loan design itself responds to saleability concerns.
- **Covenants as liquidity technology.** The paper reframes covenants not only as borrower-lender protection devices, but also as a way to make opaque loans tradable to outside buyers.
- **Hand-matched secondary-market data.** The paper builds a loan-level dataset by matching LSTA secondary-market quotations to DealScan contracts, borrower financials, lender identities, and ratings. This allows it to observe which primary loans later trade.
- **Net-worth covenant tightness.** It goes beyond counting covenants by examining net-worth slack, a contract-specific measure of covenant restrictiveness. This distinguishes “more covenants” from “tighter covenants.”
- **Borrower-side benefit of loan sales.** The paper asks why borrowers would tolerate tighter covenants and more dispersed creditor ownership. It shows that sold-loan borrowers receive unusually large loan financing around the origination year.
- **Relationship-banking contribution.** The paper contributes to relationship lending by showing that loan sales can strengthen rather than weaken future lending ties, especially for high-risk borrowers.
- **Risk-management view of loan sales.** Instead of treating sales as pure adverse selection or relationship breakdown, the paper shows that loan sales can be part of a bank's risk-management strategy that supports future credit provision.

- **Bridge between banking and market finance.** The paper explains how a market for bank loans can coexist with bank monitoring: contractual restrictions make loans more marketable while relationship banks continue to matter.

3 Data and measurement

3.1 Secondary loan-market data

- Sold loans are identified using the Loan Syndications and Trading Association (LSTA) Mark-to-Market Pricing database.
- The database contains daily secondary-market loan quotations from LPC/LSTA dealer relationships and covers a large share of U.S. secondary loan trading volume.
- A loan is classified as sold if it appears in the secondary-market quote database after origination.
- The sample period is January 1999 to December 2004.

Interpretation: The key innovation is that the authors observe ex post tradability of specific loan facilities, not just whether a loan is syndicated at origination.

3.2 Loan contracts and borrower data

- DealScan provides loan facility characteristics: loan size, maturity, loan type, loan purpose, syndication, and lead lender identity.
- Covenant information is gathered at the facility level.
- Borrower accounting information comes from Compustat.
- Borrower ratings and rating disagreement are constructed from Standard & Poor's and Moody's ratings.

Interpretation: The empirical unit is a loan facility, but the paper also constructs borrower-year panels to study access to debt capital and future relationship borrowing.

3.3 Main samples

- The full sample contains 7,261 loan facilities.
- The net-worth sample contains 2,674 facilities with minimum tangible net-worth or net-worth covenants.
- Table 1 shows that the net-worth sample is smaller and somewhat more information problematic, which makes it useful for analyzing covenant tightness.
- Table 2 shows that sold loans differ sharply from unsold loans: sold loans are larger, longer maturity, more likely syndicated, more likely term loans, and more likely to finance acquisitions or leveraged buyouts.

Interpretation: The paper does not assume sold and unsold loans are comparable. It explicitly documents differences and then uses controls, fixed effects, matching, and interactions to interpret the patterns.

3.4 Covenant measures

- **Number of covenants** counts contractual restrictions in the loan facility.
- **Net-worth slack** measures how close the borrower is to the net-worth covenant threshold. Lower slack indicates a more restrictive covenant.
- The paper interprets more covenants and lower slack as more restrictive contracting.
- Covenant restrictions are costly for borrowers because they constrain financial flexibility and can trigger technical default and renegotiation.

Interpretation: Covenants have two roles. Ex ante, they discipline borrower behavior and signal willingness to stay in a safe region. Ex post, covenant violations give lenders intervention rights.

3.5 Information and agency problem proxies

- **Common rating** equals one if Moody's and S&P agree on the borrower's long-term debt rating.
- Rating disagreement is interpreted as a proxy for greater opacity or information disagreement.
- **Lead lender in top 10** is based on yearly loan-market share and proxies for lender reputation.
- Distance-to-default proxies for borrower risk and is used heavily in the relationship-lending analysis.

Interpretation: The key prediction is that covenants should matter more when buyers face worse information problems or weaker seller reputation.

4 Models and empirical specifications

4.1 Probability of loan sale

The baseline sale equation can be written as:

$$Sold_i = \mathbf{1}\{\alpha + \beta_1 Covenants_i + \beta_2 Slack_i + \gamma' X_i + FE_i + \varepsilon_i > 0\}.$$

- $Sold_i$ equals one if loan facility i is later quoted for trade in the secondary market.
- $Covenants_i$ is the number of covenants.
- $Slack_i$ is net-worth covenant slack.
- X_i includes borrower, loan, lender, rating, industry, year, loan purpose, and loan type controls.

Interpretation: A positive coefficient on the number of covenants and a negative coefficient on slack mean that more restrictive contracts are associated with higher loan sale probability.

4.2 Restricted samples: term loans and syndicated loans

The paper re-estimates the sale model using:

- only term loans, because credit lines are harder to sell since buyers may need to provide future funding;
- only syndicated loans, because syndicated loans are already partly distributed at origination.

Interpretation: If covenants predict sales even in these restricted samples, the result is not driven only by loan type or by the difference between bilateral and syndicated facilities.

4.3 Information-problem interactions

The interaction specification is:

$$Sold_i = \mathbf{1}\{\alpha + \beta Slack_i + \theta Z_i \times Slack_i + \delta Covenants_i + \gamma' X_i + FE_i + \varepsilon_i > 0\},$$

where Z_i is either common rating or lead-lender reputation.

Interpretation: If tight covenants are less important when ratings agree or when the lead lender is reputable, then covenants are especially valuable in sale transactions with greater information problems.

4.4 Borrower access to debt capital

For matched sold and nonsold borrowers, the paper compares debt around the sold-loan year:

$$\Delta DebtMeasure_{it} = DebtMeasure_{it} - DebtMeasure_{i,t-1},$$

then contrasts sold borrowers with matched nonsold borrowers.

- Outcomes include debt-to-assets, interest-bearing debt, and loans received.
- Matching uses the year of borrowing, industry, and asset size.
- The key comparison is the difference-in-differences around the year of the sold loan.

Interpretation: This analysis addresses the borrower-side question: why accept tighter contracts? The answer is that loan sales are associated with larger debt financing exactly when the sold loan is originated.

4.5 Future lending relationship model

The paper uses a nested logit structure:

- first level: borrow again versus not borrow again;
- second level conditional on borrowing again: keep the same lead lender versus switch lead lender.

The latent utility for alternative (k, l) can be represented as:

$$U_{kl} = Y'_{kl}\alpha + Z'_{kl}\beta + \varepsilon_{kl}.$$

Interpretation: The nested structure separates the extensive margin of future borrowing from the relationship margin of keeping the same lead lender.

5 Identification and empirical design

5.1 Core identification challenge

Sold loans are not randomly selected. They are larger, riskier, more syndicated, more likely term loans, and more likely to finance acquisitions and LBOs. Therefore, the core empirical challenge is separating the role of loan saleability from borrower and loan selection.

5.2 How the paper addresses selection

- Logit models control for borrower risk, borrower size, profitability, rating status, loan size, maturity, syndication, lender reputation, loan type, loan purpose, industry, year, and lead lender fixed effects.
- The term-loan and syndicated-loan subsample tests reduce concern that results are driven by loan type or distribution at origination.
- The interaction tests use cross-sectional variation in information problems: common ratings and lead-lender reputation.
- The debt-capital analysis uses matched borrowers and difference-in-differences around the sold-loan year.
- The relationship analysis uses a nested logit to separate future borrowing from retention of the lead lender.

Interpretation: The design is not a randomized experiment. Its strength is that several independent patterns point to the same mechanism: saleable loans are more restrictively contracted, funding rises at sale origination, and relationships persist.

5.3 Main remaining limitations

- The paper generally observes that a loan trades, not necessarily the identity of the exact seller and buyer in each transaction.
- Sold loans may differ in unobserved ways that are correlated with covenant design.
- The debt-capital increase may partly reflect investment demand, such as acquisitions or LBOs, rather than pure credit supply.
- Lending relationship durability may be affected by unobserved borrower quality or lender strategy.

Interpretation: The evidence is best read as strong evidence of an equilibrium relationship between loan sales, contracting, credit availability, and relationship durability rather than a single clean causal estimate.

6 Section-by-section study guide

- **Introduction.** Sets up the puzzle: loan sales provide liquidity and risk management, but they may worsen information and relationship problems.
- **Theory and hypotheses.** Explains why covenants can mitigate buyer concerns and why borrowers may accept them if sales expand credit access.
- **Data.** Builds a facility-level sample matching LSTA secondary loan quotations to DealScan contracts and Compustat borrower information.
- **Loan contracting.** Tests whether more restrictive covenants predict loan sale probability and whether the relation is stronger under information problems.
- **Borrower debt capital.** Uses matched borrowers to show that sold-loan borrowers experience a temporary debt increase around the sale year.
- **Lending relationships.** Shows that sold borrowers are more likely to borrow again and keep the same lead lender, especially when riskier.
- **Conclusion.** Interprets loan sales as a mechanism that can support, not destroy, relationship lending.

7 Tables and figures

7.1 Figure 1: Loan sales market volume + Table 1: Loan sample summary statistics

Read:

- Figure 1 shows secondary loan trading growing from roughly 8 billion in 1991 to 238.6 billion in 2006.
- The figure establishes that loan sales are not a niche phenomenon by the mid-2000s.
- Table 1 reports 7,261 full-sample facilities and 2,674 net-worth-sample facilities.
- Only 14.8% of full-sample loans and 7.6% of net-worth-sample loans are sold.
- The net-worth sample has more covenants, smaller borrowers, fewer ratings, and more loans by lower-market-share lead lenders.

Detailed interpretation: The figure motivates the economic importance of the market. The table clarifies that the paper's covenant-tightness analysis uses a more restricted sample in which net-worth covenants are measurable. This sample is especially useful because the loans look more information sensitive.

Economic message: Loan sales became large enough to matter for corporate credit, and the loans in which covenant tightness can be measured are exactly the ones where informational contracting issues should be salient.

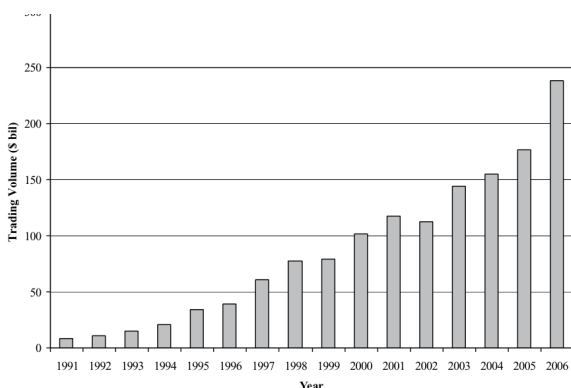


Figure 1
Loan sales market volume
This chart displays trading volume (in billions of dollars) in the secondary loan sales market from 1991 through 2006. Source: Reuters/Loan Pricing Corporation Traders Survey.

Figure 1: Loan sales market volume

Table 1
Loan sample summary statistics

Variable	Full Sample	Net Worth Sample
	Mean/median or percentage	Mean/median or percentage
Loan is Sold (0,1)	0.148	0.076
Net Worth Slack		0.132/0.101
# of Covenants	4.589/4.000	5.010/5.000
Borrower Characteristics		
Assets	2587.080/547.622	1534.996/282.691
Net Income-to-Assets	0.002/0.007	0.003/0.007
Debt-to-Assets	0.324/0.308	0.280/0.263
Distance-to-Default	2.304/1.929	2.110/1.844
Has Credit Rating (0,1)	0.460	0.310
Investment-Grade Rated (0,1)	0.192	0.121
Junk Rated (0,1)	0.268	0.189
Loan Characteristics		
Loan Size (\$ mil)	202.827/85.925	133.626/45.764
Loan Maturity (months)	40.616/36.033	37.587/36.033
Loan is Syndicated (0,1)	0.737	0.643
Lead Lender in Top 10 (0,1)	0.635	0.556
Loan Type		
Credit Line (0,1)	0.668	0.712
Term Loan (0,1)	0.295	0.263
Other Loan Type (0,1)	0.037	0.026
Loan Purpose		
Acquisition Purpose (0,1)	0.164	0.140
General Purpose (0,1)	0.463	0.509
Other Purpose (0,1)	0.097	0.052
Recapitalization Purpose (0,1)	0.210	0.252
LBO Purpose (0,1)	0.020	0.001
Miscellaneous Purpose (0,1)	0.047	0.046
Number of Loans	7261	2674

Table 1: Loan sample summary statistics

7.2 Table 2: Univariate analysis - sold loans versus not sold loans + Table 3: Probability of selling loans in the secondary market

Read:

- Table 2 shows that sold loans have lower net-worth slack, more covenants, larger borrowers, more leverage, lower distance-to-default, more ratings, larger loan size, longer maturity, and more syndication.
- Sold loans average 6.34 covenants compared with 4.28 for unsold loans.

- Sold loans are much more likely to be term loans and to finance acquisitions or LBOs.
- Table 3 shows that, conditional on controls and fixed effects, more covenants predict a higher probability of sale and lower net-worth slack predicts sale in the net-worth sample.
- Larger loans, longer maturity, syndication, borrower leverage, junk rating, and lower distance-to-default also predict sale.

Detailed interpretation: Table 2 documents strong selection into loan sales. Table 3 shows that the covenant patterns survive multivariate controls. Together, the tables demonstrate that sold loans are riskier and more complex, but their contracts are also systematically more restrictive.

Economic message: Buyers of loans appear to demand contractual safeguards. Loan liquidity is associated with more monitoring and tighter contracting, not weaker lender control.

Table 2
Univariate analysis: sold loans versus not sold loans

Variable	Loan Sale	No Loan Sale	Difference	T-ratio or Z-ratio
	Mean/percentage	Mean/percentage		
Net Worth Slack ¹	0.07	0.14	-0.07	-9.53***
# of Covenants	6.34	4.28	2.06	27.48***
Borrower Characteristics				
Assets	3981.39	2344.78	1636.61	8.16***
Net Income-to-Assets	-0.002	0.003	-0.005	-5.86***
Debt-to-Assets	0.47	0.30	0.17	20.08***
Distance-to-Default	2.13	2.33	-0.20	-4.82***
Has Credit Rating (0,1)	0.88	0.39	0.49	29.82***
Investment-Grade Rated (0,1) ²	0.12	0.53	0.41	23.41***
Junk Rated (0,1) ²	0.88	0.47	0.41	23.41***
Loan Characteristics				
Loan Size (\$ mil)	329.13	180.88	148.25	11.77***
Loan Maturity (months)	62.21	36.86	25.35	34.74***
Loan is Syndicated (0,1)	0.99	0.69	0.29	20.09***
Lead Lender in Top 10 (0,1)	0.82	0.60	0.22	13.79***
Loan Type				
Credit Line (0,1)	0.34	0.73	-0.39	-24.90***
Term Loan (0,1)	0.64	0.24	0.40	26.86***
Other Loan Type (0,1)	0.02	0.04	-0.02	-2.77***
Loan Purpose				
Acquisition Purpose (0,1)	0.27	0.15	0.12	9.98***
General Purpose (0,1)	0.36	0.48	-0.12	-7.09***
LBO Purpose (0,1)	0.05	0.01	0.03	7.15***
Other Purpose (0,1)	0.04	0.11	-0.06	-6.58***
Recapitalization Purpose (0,1)	0.22	0.21	0.01	0.87
Miscellaneous Purpose (0,1)	0.06	0.04	0.01	2.08**
Number of Loans	1075	6186		

This table tests for differences in means or percentages between loans that are sold and loans that are not sold.

Table 3
Probability of selling loans in the secondary market

	Net Worth Sample		Full Sample	
	(1)	(2)	(3)	(4)
Net Worth Slack	-2.436** (1.94)	-2.623* (1.84)		
# of Covenants	0.357*** (5.17)	0.337*** (4.21)	0.277*** (8.54)	0.252*** (7.50)
Log of Assets	0.702*** (4.25)	0.651*** (3.49)	0.481*** (5.96)	0.424*** (5.02)
Net Income-to-Assets	0.823 (0.13)	-4.612 (0.69)	-8.078*** (2.99)	-8.605*** (3.14)
Debt-to-Assets	0.682* (1.81)	0.683 (0.75)	0.981*** (3.85)	1.352*** (4.70)
Distance-to-Default	-0.011 (0.09)	-0.036 (0.27)	-0.186*** (3.44)	-0.198*** (3.55)
Junk Rated (0,1)	1.460*** (3.92)	1.362*** (3.17)	1.260*** (7.56)	1.101*** (6.47)
Investment-Grade Rated (0,1)	0.094 (0.19)	-0.054 (0.10)	-0.079 (0.31)	-0.100 (0.40)
Log of Loan Size	0.857*** (4.64)	0.945*** (4.53)	0.608*** (7.77)	0.604*** (7.44)
Log of Loan Maturity	1.500*** (5.68)	1.728*** (6.00)	1.362*** (10.14)	1.375*** (10.62)
Loan is Syndicated (0,1)	1.321* (1.94)	1.004 (1.19)	1.465*** (4.09)	1.238*** (3.45)
Lead Lender in Top 10 (0,1)	0.134 (0.39)	-0.557 (1.02)	0.275 (1.60)	0.515** (2.11)
Constant	-43.376*** (11.00)	-43.814*** (10.02)	-31.987*** (18.75)	-28.910*** (15.32)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Loan Purpose Fixed Effects	Yes	Yes	Yes	Yes
Loan Type Fixed Effects	Yes	Yes	Yes	Yes
Lead Lender Fixed Effects	No	Yes	No	Yes
Log Likelihood	-318.56	-280.39	-1506.52	-1409.02
Pseudo R-squared	0.56	0.56	0.51	0.51
Number of Loans	2674	1897	7261	6213

This table presents results of logit models. The dependent variable is an indicator variable that is 1 if the loan is sold and 0 if the loan is not sold. The independent variables are described in the Appendix. The Net Worth Sample, which includes loans that have a minimum (tangible) net worth covenant, is used in estimations displayed in columns 1 and 2. Standard errors are clustered at the loan deal level, and T-ratios are in parentheses. ***, **, * indicates statistically different than zero at the 1%, 5%, and 10% level, respectively.

Table 2: Univariate analysis - sold loans versus not sold loans

Table 3: Probability of selling loans in the secondary market

7.3 Table 4: Probability of selling loans restricted to term loans + Table 5: Sold loans - timing, pricing, and performance

Read:

- Table 4 restricts the sample to term loans and finds the same core pattern: lower slack and more covenants predict sale.
- The covenant-count coefficients remain positive and significant, showing that the result is not only due to credit lines being harder to sell.
- Table 5 shows that 61.77% of sold loans first trade within the first month after origination and 87.71% within the first year.

- Median first-sale price is 99.78% of par; mean first-sale price is 97.86% of par.
- 93.50% of sold loans have no rating change between origination and first sale; only 5.75% are downgraded.
- Last-sale timing is much more spread out, indicating that many loans continue to trade long after origination.

Detailed interpretation: Table 4 confirms robustness on the facility type most naturally suited to trading. Table 5 rules out the idea that the sample is mainly distressed dumping: sales occur quickly after origination, prices are close to par, and rating changes are usually absent.

Economic message: Loan sales are part of the original financing and distribution model, not primarily a post-origination distress response.

Table 4
Probability of selling loans in the secondary market sample restricted to term loans

	Net Worth Sample		Full Sample	
	(1)	(2)	(3)	(4)
Net Worth Slack	-3.434* (1.70)	-3.992* (1.75)		
# of Covenants	0.206* (1.92)	0.227** (2.04)	0.138*** (3.09)	0.116*** (2.93)
Log of Assets	0.372 (1.43)	0.500 (1.58)	0.215* (1.82)	0.221** (2.16)
Net Income-to-Assets	0.922 (0.09)	7.595 (0.57)	-2.846 (0.75)	-2.664 (0.74)
Debt-to-Assets	-0.305 (0.30)	-0.224 (0.19)	0.742*** (3.75)	1.002*** (3.31)
Distance-to-Default	-0.072 (0.39)	-0.081 (0.37)	-0.203** (2.53)	-0.262*** (3.26)
Junk Rated (0,1)	2.039*** (4.07)	1.810*** (3.67)	1.385*** (6.62)	1.285*** (6.68)
Investment-Grade Rated (0,1)	-0.019 (0.02)	-0.344 (0.37)	0.022 (0.06)	0.047 (0.14)
Log of Loan Size	1.386*** (4.46)	1.403*** (4.14)	0.971*** (7.74)	0.906*** (7.91)
Log of Loan Maturity	1.586*** (3.49)	1.878*** (3.72)	1.328*** (6.11)	1.432*** (7.82)
Loan is Syndicated (0,1)	0.712 (0.83)	0.512 (0.46)	1.589*** (4.05)	1.412*** (3.55)
Lead Lender in Top 10 (0,1)	0.757 (1.59)	-0.316 (0.33)	0.415** (2.03)	0.310 (0.49)
Constant	-42.462*** (7.12)	-44.356*** (6.99)	-30.371*** (13.21)	-27.093*** (3.73)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Loan Purpose Fixed Effects	Yes	Yes	Yes	Yes
Lead Lender Fixed Effects	No	Yes	No	Yes
Log Likelihood	-127.51	-111.91	-637.68	-587.70
Pseudo R-squared	0.62	0.59	0.53	0.49
Number of Loans	702	470	2143	1719

This table presents results of logit models. Only term loans are used when estimating these models. The dependent variable is an indicator variable that is 1 if the loan is sold and 0 if the loan is not sold. The independent variables are described in the Appendix. The Net Worth Sample, which includes loans that have a minimum (tangible) net worth covenant, is used in estimations displayed in columns 1 and 2. Standard errors are clustered at the loan

Table 5
Sold loans: timing, pricing, and performance

Panel A: Time to first sale

	Number of loans sold	Cumulative percentage
1st month	593	61.77
2nd month	52	67.19
3rd month	35	70.83
4th to 6th month	80	79.17
7th to 12th month	82	87.71
2nd year	75	95.52
3rd year	23	97.92
4th year	15	99.48
5th year and longer	5	100.00

Panel B: Price at first sale

Distribution percentile	1%	5%	10%	25%	50%	75%	90%	95%	99%
Price (% of par)	58.25	87.25	95.25	98.78	99.78	100.44	101.00	101.25	101.94
	Mean deviation								
	97.86	7.91							

Panel C: Performance of sold loans between origination and sale

Ratings changes	Percentage of sold loans	Distance-to-default changes
No rating change	93.50	Median 0.00
Upgrade	0.75	Mean 0.068
Downgrade	5.75	Standard deviation 0.706

Panel D: Time to last sale

	Number of loans sold	Cumulative percentage
1st to 3rd month	6	0.63
4th to 6th month	22	2.29
7th to 12th month	142	17.71
13th to 18th month	171	35.52
19th to 24th month	141	50.21
3rd year	172	68.13
4th year	139	82.60
5th year	116	94.69
6th year and longer	51	100.00

Panel A displays the time between the loan origination date and the first date the loan is quoted for trade. Panel B provides the distribution of prices for sold loans at the first date the loan is quoted for trade. The price is the midpoint of the mean bid price and mean ask price and is quoted as a percentage of par. Panel C displays the performance of loans between the origination date and the first date the loan is quoted for trade. An upgrade (downgrade) (no rating change) occurs if the borrower's Standard & Poor's long-term debt rating improves (declines) (does not change) between origination and sale. The Standard & Poor's long-term debt rating is computed quarterly and is based on the following scale (from highest quality to lowest): AAA, AA, A, BBB, ...

Table 4: Probability of selling loans restricted to term loans

Table 5: Sold loans - timing, pricing, and performance

7.4 Table 6: Probability of selling loans restricted to syndicated loans + Table 7: Credit rating disagreement and lender reputation

Read:

- Table 6 restricts the sample to syndicated loans and still finds that more covenants and lower slack predict sale.

- This matters because syndication itself already distributes the loan among multiple lenders at origination.
- Table 7 interacts covenant slack and covenant count with common rating and lead lender reputation.
- Net-worth slack remains strongly negative, meaning tighter covenants increase sale probability.
- The common-rating interaction and lead-lender-reputation interaction suggest that tight covenants matter more when information problems are more severe.

Detailed interpretation: Tables 6 and 7 strengthen the contracting interpretation. If sold-loan covenants only reflected syndication mechanics, the result should disappear in the syndicated subsample. It does not. The interaction evidence then shows that the covenant-sale relation is connected to information problems rather than pure loan size.

Economic message: Covenants appear to substitute for missing information and trust in the secondary market. When uncertainty is greater, saleable loans require tighter contractual protection.

Table 6
Probability of selling loans in the secondary market sample restricted to syndicated loans

	Net Worth Sample		Full Sample	
	(1)	(2)	(3)	(4)
Net Worth Slack	-2.532*	-2.914**		
	(1.93)	(1.97)		
# of Covenants	0.344***	0.336***	0.266***	0.245***
	(4.92)	(4.28)	(8.12)	(7.19)
Log of Assets	0.693***	0.618***	0.475***	0.417***
	(4.15)	(3.33)	(5.70)	(4.77)
Net Income-to-Assets	0.155	-6.129	-8.577***	-9.166***
	(0.02)	(0.93)	(3.10)	(3.28)
Debt-to-Assets	0.706	0.611	1.467***	1.343***
	(0.90)	(0.65)	(5.56)	(4.63)
Distance-to-Default	-0.005	-0.026	-0.153***	-0.172***
	(0.04)	(0.19)	(2.87)	(3.11)
Junk Rated (0,1)	1.447***	1.382***	1.204***	1.101***
	(3.84)	(3.25)	(7.18)	(6.37)
Investment-Grade Rated (0,1)	0.053	-0.072	-0.112	-0.130
	(0.11)	(0.13)	(0.44)	(0.51)
Log of Loan Size	0.834***	0.958***	0.608***	0.613***
	(4.57)	(4.63)	(7.53)	(7.31)
Log of Loan Maturity	1.473***	1.702***	1.421***	1.422***
	(5.57)	(5.89)	(10.30)	(10.85)
Lead Lender in Top 10 (0,1)	0.154	-0.507	0.239	0.457*
	(0.44)	(0.95)	(1.36)	(1.84)
Constant	-41.377***	-42.249***	-30.687***	-27.753***
	(10.29)	(9.37)	(17.69)	(14.52)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Loan Purpose Fixed Effects	Yes	Yes	Yes	Yes
Loan Type Fixed Effects	Yes	Yes	Yes	Yes
Lead Lender Fixed Effects	No	Yes	No	Yes
Log Likelihood	-314.83	-274.79	-1452.45	-1367.76
Pseudo R-squared	0.49	0.53	0.45	0.47
Number of Loans	1720	1466	5352	5033

This table presents results of logit models. Only loans that are syndicated are used when estimating these models. The dependent variable is an indicator variable that is 1 if the loan is sold and 0 if the loan is not sold. The independent variables are described in the Appendix. The Net Worth Sample, which includes loans that have a minimum (tangible) net worth covenant, is used in estimations displayed in columns 1 and 2. Standard errors are clustered at the loan deal level, and T-ratios are in parentheses. ***, **, * indicate significantly different than zero at the 1%, 5%, and 10% levels, respectively.

Table 7
Probability of selling loans: credit rating disagreement and lender reputation

	(1)	(2)	(3)	(4)
Net Worth Slack	-6.208***	-5.723**	-6.616***	-8.233**
	(3.15)	(2.33)	(2.71)	(0.82)
Common Rating*	4.496**	4.861*		
	(1.96)	(1.72)		
Net Worth Slack			5.092*	6.991*
Lead Lender in Top 10*			(1.81)	(1.73)
Net Worth Slack			0.527***	0.463***
# of Covenants	0.392***	0.466***	(4.25)	(2.57)
	(2.99)	(2.99)		
Common Rating*	-0.093	-0.215		
# of Covenants	(0.70)	(1.31)		
Lead Lender in Top 10*			-0.200	-0.155
# of Covenants			(1.50)	(0.82)
Log of Assets	0.795***	0.784***	0.730***	0.670***
	(4.44)	(3.76)	(4.39)	(3.59)
Net Income-to-Assets	-3.761	-4.831	0.185	-4.896
	(0.50)	(0.56)	(0.03)	(0.76)
Debt-to-Assets	0.803	0.610	0.606	0.625
	(0.95)	(0.64)	(1.14)	(0.67)
Distance-to-Default	0.038	0.028	-0.010	-0.046
	(0.28)	(0.21)	(0.08)	(0.35)
Junk Rated (0,1)	1.334***	1.509***	1.387***	1.387***
	(3.29)	(3.59)	(3.77)	(3.34)
Investment-Grade Rated (0,1)			0.002	-0.028
			(0.00)	(0.05)
Log of Loan Size	0.752***	0.937***	0.850***	0.960***
	(3.38)	(3.57)	(4.61)	(4.63)
Log of Loan Maturity	1.701***	1.858***	1.451***	1.639***
	(5.45)	(5.53)	(5.42)	(5.88)
Loan is Syndicated (0,1)	0.906	0.520	1.165*	0.798
	(1.26)	(0.64)	(1.80)	(0.96)
Lead Lender in Top 10 (0,1)	0.102	-0.219	1.171	0.372
	(0.26)	(0.35)	(1.14)	(0.27)
Common Rating (0,1)	-0.049	0.439		
	(0.04)	(0.40)		
Constant	-44.665***	-45.502***	-46.061***	-44.908***
	(9.42)	(8.06)	(11.04)	(9.77)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Loan Purpose Fixed Effects	Yes	Yes	Yes	Yes
Loan Type Fixed Effects	Yes	Yes	Yes	Yes
Lead Lender Fixed Effects	No	Yes	No	Yes
Log Likelihood	-239.25	-210.15	-314.76	-281.93
Pseudo R-squared	0.45	0.49	0.56	0.56
Number of Loans	829	750	2674	1901

This table presents results of logit models. The dependent variable is an indicator variable that is 1 if the loan is sold and 0 if the loan is not sold. The independent variables are described in the Appendix. The Net Worth Sample, which includes loans that have a minimum (tangible) net worth covenant, is used in all estimations. "Common Rating" indicates that Moody's and Standard & Poor's have the same long-term debt ratings. Since the variable is not coded for borrowers that are unrated by Moody's or Standard & Poor's, the models in columns 1 and 2 are estimated for loans where the borrower has a credit rating. "Lead Lender in Top 10" indicates that a lead lender has a loan market share among the highest 10 lenders during the year prior to the loan. Standard errors are clustered at the loan deal level, and T-ratios are in parentheses. ***, **, * indicate significantly different than zero at the 1%, 5%, and 10% levels, respectively.

Table 6: Probability of selling loans restricted to syndicated loans

Table 7: Probability of selling loans - credit rating disagreement and lender reputation

7.5 Table 8: Loan sales and debt capital + Table 9: Probability of keeping the same lead lender

Read:

- Table 8 compares sold borrowers with matched nonsold borrowers around the year of the sold loan.
- In the sold-loan year, sold borrowers have higher debt-to-assets, higher interest-bearing debt, and much larger loans received.
- Loans received rise from roughly 433 million one year before sale to about 887 million in the sold-loan year.
- The year-of-sale difference-in-differences estimates are significant, while pre-sale changes are not.
- Table 9 shows that 84.74% of sold-loan borrowers receive another loan, compared with 68.75% for nonsold borrowers.
- Conditional on receiving a future loan, 83.53% of sold-loan borrowers keep the same lead lender, compared with 75.83% for nonsold borrowers.
- The relationship effect is strongest for low-distance-to-default borrowers, where loan sales may be most useful for credit-risk management.

Detailed interpretation: Table 8 provides the borrower benefit: loan sales coincide with a temporary increase in private debt availability. Table 9 answers the concern that selling loans destroys relationships. The evidence goes the other way: sold borrowers keep borrowing and often retain the same lead lender.

Economic message: Borrowers accept tighter covenants because loan sales relax funding constraints and help banks keep lending to risky relationship borrowers.

Table 8
Loan sales and debt capital

	(1)	(2)	(3)	(4)
	Two years before sold loan	One year before sold loan	Year of sold loan	One year after sold loan
Debt-to-Assets				
Sold Borrower	0.451	0.459	0.476	0.452
Matched Not Sold Borrower	0.308	0.311	0.314	0.316
p-value for differences in means	0.000	0.000	0.000	0.000
p-value for differences-in-differences (year t versus $t - 1$)		0.570	0.009	0.012
Interest-Bearing Debt				
Sold Borrower	1573.231	1661.997	1857.745	1704.480
Matched Not Sold Borrower	1105.061	1128.343	1146.257	1222.903
p-value for differences in means	0.002	0.000	0.000	0.000
p-value for differences-in-differences (year t versus $t - 1$)		0.117	0.009	0.005
Loans Received (in millions)				
Sold Borrower	447.594	433.296	886.691	306.563
Matched Not Sold Borrower	344.743	281.259	396.260	274.260
p-value for differences in means	0.160	0.000	0.000	0.330
p-value for differences-in-differences (year t versus $t - 1$)		0.526	0.000	0.000

This table provides means of debt characteristics for a sample of borrowers, based on whether their loans are sold. Borrowers are separated into two samples: "Sold Borrowers" have a loan sold at some time between 1999 and 2004; "Not Sold Borrowers" never have a loan sold between 1999 and 2004. For each sold borrower, we identify the origination fiscal year of the sold loan. A matching not sold borrower is found using the following method: (1) *Year of Borrowing*: has a loan in the same fiscal year as sold loan; (2) *Industry*: has the same two-digit SIC Code. If there is more than one match, then we choose based on (3) *Asset Size*: the not sold borrower with closest level of assets. There are 520 matched borrowers. The matched borrowers are tracked for two years before and one year after the matching year ("Year of the Sold Loan"). The unit of observation is the borrower-fiscal year. Compustat Variables: Interest-Bearing Debt is long-term debt plus debt in current liabilities; Debt-to-Assets is interest-bearing debt divided by book assets. *DealScan* Variable: Private Debt Issuance is total notional amount of loans. Dollar values are expressed in January 1999 dollars. "Sold Borrower" and "Matched Not Sold Borrowers" provide means for the matched borrower samples during the year specified in the column heading. The results of

Table 8: Loan sales and debt capital

Table 9
Univariate analysis: probability of keeping the same lead lender

Panel A: Percentage that receive another loan

	Loan sale percentage	No loan sale percentage	Difference (%)	Proportion test Z-ratio
Full Sample	84.74	68.75	15.99	10.68***
Low Distance-to-Default	82.22	62.69	19.53	7.57***
Mid Distance-to-Default	86.01	67.53	18.47	6.93***
High Distance-to-Default	86.54	75.80	10.74	4.31***

Panel B: Conditional on receiving another loan, percentage that keep the same lead lender

	83.53	75.83	7.70	5.03***
Full Sample				
Low Distance-to-Default	79.58	64.10	15.48	5.36***
Mid Distance-to-Default	84.75	75.11	9.64	3.57***
High Distance-to-Default	85.80	86.93	-1.12	-0.50

Panel A provides the percentage of borrowers that receive another loan between the loan date and December 31, 2005, split on whether the original loan was sold. For those borrowers that receive at least one future loan, panel B provides the percentage of borrowers that borrow again from the same lead lender on at least one future loan, split on whether the initial loan was sold. "Low (Mid, High) Distance-to-Default" indicates that the borrower's distance-to-default is in the lowest (middle, highest) one-third of the full sample. Z-ratios for the difference in proportions are provided in the last column. ***, **, * indicates significance different than zero at the 1%, 5%, and 10% level, respectively.

Table 9: Probability of keeping the same lead lender

7.6 Figure 2: Nesting structure + Table 10: Nested logit models of keeping the same lead lender

Read:

- Figure 2 shows the nested logit structure: first the borrower either borrows again or does not; conditional on borrowing again, the borrower either keeps or does not keep the same lead lender.
- This structure separates future access to credit from relationship durability.

- Table 10 shows that loan sales are positively associated with keeping the same lead lender after controlling for borrower and lender characteristics.
- In the interaction specification, the positive effect is strongest for low-distance-to-default borrowers.
- The high-distance-to-default interaction is negative, implying that the loan sale relationship effect is not present in the same way for safer borrowers.
- Prior lending relationships and top-10 lead lenders strongly predict keeping the same lender.

Detailed interpretation: Figure 2 clarifies the econometric design, while Table 10 shows that the relationship result is not merely driven by sold borrowers being more likely to borrow again. The positive coefficient on loan sales in the keep-lender equation means that among the relevant future-borrowing decisions, sold loans are associated with continued lead-lender relationships.

Economic message: Loan sales can be relationship-preserving because they help lead banks manage exposure while continuing to serve risky borrowers over time.

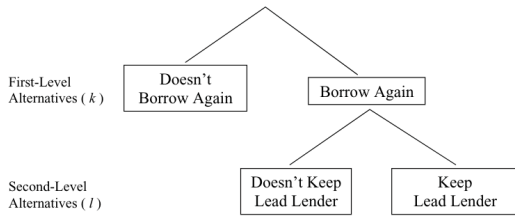


Figure 2
Nesting structure
This figure presents the nesting structure for the nested logit model. Each borrower has a first-level alternative (Doesn't Borrow Again, Borrow Again) and a second-level alternative (Doesn't Keep Lead Lender, Keep Lead Lender).

Figure 2: Nesting structure for the nested logit model

Table 10
Multivariate models: probability of keeping the same lead lender

	(1)	(2)
"Keep Lender" versus "Doesn't Keep Lender"		
Loan is Sold (0,1)	0.223** (2.20)	0.523*** (3.39)
Mid Distance-to-Default * Loan is Sold		-0.185 (0.79)
High Distance-to-Default * Loan is Sold		-0.610** (2.39)
Prior Lending Relationship (0,1)	0.772*** (10.71)	0.332*** (3.11)
Mid Distance-to-Default * Prior Lending Relationship		0.501*** (3.44)
High Distance-to-Default * Prior Lending Relationship		0.839*** (5.12)
Lead Lender in Top 10	1.118*** (15.32)	0.773*** (7.17)
Mid Distance-to-Default * Lead Lender in Top 10		0.235 (1.60)
High Distance-to-Default * Lead Lender in Top 10		0.768*** (4.69)
Constant	0.003 (0.03)	0.010 (0.09)
Industry Fixed Effects	Yes	Yes
"Borrow Again" versus "Doesn't Borrow Again"		
Log of Assets	0.427*** (16.03)	0.427*** (15.87)
Net Income-to-Assets	4.060*** (4.03)	3.507*** (3.45)
Debt-to-Assets	0.367*** (3.11)	0.358*** (2.99)
Distance-to-Default	-0.071*** (3.02)	
Mid Distance-to-Default (0,1)		0.024 (0.22)
High Distance-to-Default (0,1)		0.367*** (2.68)
Junk Rated (0,1)	0.149* (1.66)	0.157* (1.73)
Investment-Grade Rated (0,1)	0.380*** (3.02)	0.416*** (3.29)
Year Fixed Effects	Yes	Yes
Inclusive Value: IV (Borrow Again)	0.503**	0.516**
LR Test: IV (Borrow Again) = 1	3.18*	2.53
Log Likelihood	-6251.05	-6166.44
Number of Loans	7261	7261

This table presents results of nested logit models. The dependent variable indicates whether the issuer (1) "Doesn't Borrow Again," (2) "Borrow Again" and "Keep Lead Lender," or (3) "Borrow Again," and "Doesn't Keep Lead Lender." "Doesn't Borrow Again" indicates that the borrower does not receive any future loans. "Keep Lead Lender" indicates that the borrower receives at least one future loan and keeps an original lead lender in at least one future loan. "Doesn't Keep Lead Lender" indicates that the borrower receives at least one future loan but never keeps the same lead lender. Let "Borrow Again" and "Doesn't Borrow Again" belong to the upper nest and "Keep Lead Lender," and "Doesn't Keep Lead Lender" belong to the lower nest. Independent variables that affect outcomes in both the upper and lower nests are listed under the heading "Keep Lender versus Doesn't Keep Lender." The displayed coefficients for these variables provide the effect of the variables on keeping the same lender as opposed to not keeping the same lender. Independent variables that only affect the choice in the upper nest are listed under the heading "Borrow Again versus Doesn't Borrow Again." The displayed coefficients for these variables provide the effect on borrowing again instead of not borrowing again. The variables are described in the Appendix. In column 2, "Mid (High) Distance-to-Default" indicates that the borrower's distance-to-default is in the middle (highest) one-third of the sample. *T*-ratios are in parentheses. ***, **, * indicate significantly different than zero at the 1%, 5%, and 10% levels, respectively.

borrower risk class. We expect the positive effect of loan sales on lending

Table 10: Multivariate models - probability of keeping the same lead lender

8 Result map

- **Contracting result.** Sold loans have more covenants and tighter net-worth covenants. This holds in full, net-worth, term-loan, and syndicated-loan samples.
- **Information-problem result.** Covenant tightness matters more when ratings disagree and when the lead lender is not highly reputable.
- **Timing result.** Loan sales usually occur shortly after origination and near par, so the results are not driven mainly by distressed secondary sales.

- **Credit-availability result.** Sold borrowers receive significantly more loan financing in the origination year relative to matched nonsold borrowers.
- **Temporary-financing result.** Borrower debt falls after the sold-loan year, suggesting that loan sales help finance discrete large projects rather than permanently higher leverage.
- **Relationship result.** Sold borrowers are more likely to receive future loans and keep the same lead lender.
- **Risk-management result.** The relationship result is strongest among high-risk borrowers, consistent with loan sales letting banks manage exposure while preserving relationships.

9 Short summary

- The paper studies the rise of secondary-market loan sales using facility-level loan data.
- Loan sales create information and agency problems between borrowers, sellers, and buyers.
- Sold loans are larger, riskier, more syndicated, more likely term loans, and more likely to finance acquisitions or LBOs.
- Sold loans include more covenants and tighter net-worth covenants.
- Covenant-sale relations are stronger when information problems are more severe.
- Sold loans usually trade soon after origination and close to par, reducing concern that the sample is mainly distressed loan sales.
- Sold-loan borrowers experience a temporary increase in debt capital and loans received.
- Loan sales do not sever lending relationships; sold borrowers are more likely to keep the same lead lender.
- The relationship effect is strongest among riskier borrowers.
- The paper interprets loan sales as a credit-risk management tool that expands private credit availability while preserving relationship lending.

10 Conclusion

10.1 Core conclusion

Drucker and Puri show that secondary loan sales are associated with deliberate contract design. Sold loans are not simply loans that banks originate carelessly and then offload. They are loans with more restrictive covenants, especially when the transaction is likely to suffer from information and agency problems.

10.2 Economic intuition

A loan is difficult to sell because buyers worry about hidden borrower quality, weak monitoring, and seller moral hazard. Covenants mitigate these concerns by giving buyers observable control rights and early-warning triggers. Borrowers bear covenant costs, but in return they appear to obtain larger loans and greater access to private debt funding.

10.3 Incremental contribution

The paper's incremental contribution is to connect three margins that are often studied separately: secondary loan trading, loan contract design, and relationship lending. It shows that the development of a secondary

loan market changes primary loan contracting, increases credit availability for borrowers with large financing needs, and can preserve relationship lending through bank risk management.

10.4 Interpretation for loan-market design

The results imply that loan market liquidity depends on contractibility. A liquid loan market for risky borrowers requires mechanisms that reduce information frictions, such as covenants, monitoring rights, and lead-lender reputation. Contract design is therefore part of the market infrastructure that makes loans tradeable.

10.5 Implications for relationship banking

A common concern is that loan sales weaken bank-borrower relationships. The paper provides a more nuanced view. If loan sales let banks reduce credit exposure while keeping origination and monitoring roles, then loan sales can actually support future lending relationships, especially for high-risk borrowers that need relationship banks most.

10.6 Limitations

The evidence is observational. Sold loans differ from unsold loans, and unobserved borrower or lender characteristics could still matter. The paper also cannot always identify the exact seller and buyer, which limits direct tests of lead-lender selling. The results should therefore be read as a consistent set of empirical patterns rather than a randomized causal design.

10.7 Takeaway

The paper's main lesson is that the secondary loan market is not the opposite of relationship banking. Properly designed contracts can make relationship-originated loans tradable, and trading can help relationship lenders manage risk and continue lending.